

D288

Back-End Programming

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A mobile app for a chain of 30 fitness clubs retries requests automatically when the network drops. The team wants the retry of an update to a member's full profile to leave the same result no matter how many times it is sent. The app is rated 4.7 stars. Which HTTP method fits?

- A. GET
- B. Any method, because retries are always safe.
- C. PUT
- D. POST

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A client sends the same request twice by mistake, and the server ends up with two new order records. Which method was most likely used?

- A. POST
- B. PUT
- C. GET
- D. DELETE

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A logged-in user with the viewer role, working from the Denver office, tries to delete an invoice through the API. Only admins may delete invoices. The server runs on eight cores. Which status code should the API return?

- A. 403 Forbidden
- B. 500 Internal Server Error
- C. 401 Unauthorized
- D. 404 Not Found

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

An API successfully creates a new customer record in response to a request. Which status code best reports that outcome?

- A.** 400 Bad Request
- B.** 204 No Content
- C.** 200 OK
- D.** 201 Created

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A travel company with offices in five countries is designing an endpoint to fetch one booking by its ID, 8812. The team uses JSON for all responses. Which request follows REST conventions?

- A.** POST /bookings/8812
- B.** GET /bookings?country=5
- C.** GET /bookings/8812
- D.** GET /getBooking?id=8812

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

In a school district's app, each Teacher has many Courses, and each Course belongs to one Teacher. The district serves 18,000 students. In the relational tables, where does the foreign key go?

- A.** A course_id column in the Teacher table.
- B.** A teacher_id column in the Course table.
- C.** Keys in both tables pointing at each other.
- D.** A separate join table is always required.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

An Order table has a customer_id foreign key. What relationship does this describe?

- A.** One order has many customers.
- B.** One customer has many orders.
- C.** One-to-one, because each order has one customer_id.
- D.** Many-to-many between orders and customers.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A Spring Data JPA project for a veterinary clinic with two locations needs all pets whose species is dog. The Pet entity has a species field. A senior developer suggests writing as little code as possible. What is the simplest approach?

- A. Load every pet with findAll() and filter in the view.
- B. Add List<Pet> findBySpecies(String species) to the PetRepository interface.
- C. Write a JDBC SQL query inside the controller.
- D. Every custom query must be written as native SQL.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

An e-commerce API that runs in Docker containers must apply a 10% loyalty discount to orders over \$200. The controller receives the order as JSON. Where should the discount rule live?

- A. In the repository.
- B. In the Docker configuration.
- C. In the controller.
- D. In the service layer.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

In a controller-service-repository design, which call path is correct for saving a new order?

- A. Controller calls repository directly and skips the service.
- B. Service calls controller to get the order data.
- C. Repository calls service; service calls controller.
- D. Controller calls service; service calls repository.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A payment service debits \$500 from a customer's account and then credits a merchant, all in one method marked @Transactional. The credit fails with a runtime exception. The customer has been a member for nine years. What is the account state afterward?

- A.** The debit is saved and the credit is not.
- B.** Both are saved, because the debit succeeded first.
- C.** Neither the debit nor the credit is saved.
- D.** The debit is refused for long-time members.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

An Angular front end served at http://localhost:4200 calls a Spring API at http://localhost:8080. Both run on the same laptop, which has 16 GB of RAM. The browser blocks the call. What fixes it?

- A.** Configure the API to allow the origin http://localhost:4200 with CORS.
- B.** Add CORS settings to the front end instead of the API.
- C.** Nothing; the same host is always the same origin.
- D.** Add more RAM so the browser can handle both apps.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A mobile app ~~for a chain of 30 fitness clubs~~ retries requests automatically when the network drops. The team wants the retry of an update to a member's full profile to leave the same result no matter how many times it is sent. ~~The app is rated 4.7 stars.~~ Which HTTP method fits?

Target word: idempotent method for a full update **Decides:** Same result on repeat sends = idempotent; full replacement = PUT.

A	Adjacent category	GET reads; it must not change data.
B	Absolute language	Retries are safe only for idempotent methods.
C	Correct	PUT replaces the resource; sending it twice leaves the same state.
D	Confusion set	POST is not idempotent; repeats can create duplicates.

2. Answer A Concept 1 · transfer

A client sends the same request twice by mistake, and the server ends up with two new order records. Which method was most likely used?

Target word: reverse: infer the method from a non-idempotent result **Decides:** Duplicate creation on repeat means a non-idempotent create.

A	Correct	POST creates a new resource each time it is sent.
B	Inverted operation	PUT to the same URI replaces, so repeats leave one record.
C	Confusion set	GET is safe and should not create anything.
D	Adjacent category	DELETE removes records; it does not create them.

3. Answer A Concept 2

A logged-in user with the viewer role, ~~working from the Denver office,~~ tries to delete an invoice through the API. Only admins may delete invoices. ~~The server runs on eight cores.~~ Which status code should the API return?

Target word: status for authenticated but not permitted **Decides:** The user is known (logged in) but lacks permission.

A	Correct	Authenticated, but not authorized for this action.
B	Inverted operation	Blames the server for a client-side permission problem.
C	Confusion set	401 means not authenticated; this user is logged in.
D	Adjacent category	The invoice exists; the problem is permission.

4. Answer D Concept 2 · transfer

An API successfully creates a new customer record in response to a request. Which status code best reports that outcome?

Target word: status for a successful create **Decides:** A new resource was created.

A	Inverted operation	Reports a client error, not success.
B	Confusion set	Success with no body, typical for delete or update.
C	Adjacent category	Generic success; 201 is the specific code for creation.
D	Correct	Signals success and that a new resource exists.

5. Answer C Concept 3

A travel company ~~with offices in five countries~~ is designing an endpoint to fetch one booking by its ID, 8812. ~~The team uses JSON for all responses.~~ Which request follows REST conventions?

Target word: RESTful URI for one resource **Decides:** Noun collection plus a path ID, with GET as the action.

A	Adjacent category	POST creates or processes; it is not used to read.
B	Camouflage	Filters by the office detail instead of fetching the booking.
C	Correct	Plural noun collection and a path variable for one item.
D	Confusion set	Puts a verb in the URI; the method already says get.

6. Answer B Concept 4

In a school district's app, each Teacher has many Courses, and each Course belongs to one Teacher. ~~The district serves 18,000 students.~~ In the relational tables, where does the foreign key go?

Target word: foreign key placement in one-to-many **Decides:** Course is the many side.

A	Inverted operation	One column could hold only one course per teacher.
B	Correct	The many side holds the key pointing to its one parent.
C	Adjacent category	Duplicates the relationship and risks inconsistency.
D	Absolute language	Join tables are needed for many-to-many, not one-to-many.

7. Answer B Concept 4 · transfer

An Order table has a customer_id foreign key. What relationship does this describe?

Target word: reverse: infer the relationship from the key **Decides:** The table holding the key is the many side.

A	Inverted operation	Reads the key backward.
B	Correct	Many orders can store the same customer_id.
C	Dropped element	Ignores that many orders can share one customer.
D	Adjacent category	Many-to-many needs a join table.

8. Answer B Concept 5

A Spring Data JPA project ~~for a veterinary clinic with two locations~~ needs all pets whose species is dog. The Pet entity has a species field. ~~A senior developer suggests writing as little code as possible.~~ What is the simplest approach?

Target word: repository derived query **Decides:** The framework builds a query from a method name matching the field.

A	Confusion set	Works but pulls all rows and puts logic in the view.
B	Correct	Derived query methods are generated from the name.
C	Adjacent category	Moves data access into the wrong layer and adds code.
D	Absolute language	Derived queries cover simple filters without SQL.

9. Answer D Concept 6

An e-commerce API that runs in Docker containers must apply a 10% loyalty discount to orders over \$200. The controller receives the order as JSON. Where should the discount rule live?

Target word: which layer owns business logic **Decides:** A pricing rule is business logic.

A	Adjacent category	Repositories handle persistence, not pricing.
B	Confusion set	Deployment settings do not hold business logic.
C	Camouflage	The controller receiving JSON does not make it the place for rules.
D	Correct	The service holds business rules between controller and repository.

10. Answer D Concept 6 • transfer

In a controller-service-repository design, which call path is correct for saving a new order?

Target word: direction of calls between layers **Decides:** Each layer calls the one directly below.

A	Dropped element	Drops the layer that holds business rules.
B	Confusion set	Lower layers do not call the web layer.
C	Inverted operation	Reverses the dependency direction.
D	Correct	Top-down, one layer at a time.

11. Answer C Concept 7

A payment service debits \$500 from a customer's account and then credits a merchant, all in one method marked @Transactional. The credit fails with a runtime exception. ~~The customer has been a member for nine years.~~ What is the account state afterward?

Target word: result of a failed transaction **Decides:** All writes in the transaction roll back together.

A	Dropped element	Ignores the all-or-nothing rule.
B	Inverted operation	The failure undoes earlier writes, not the other way around.
C	Correct	The runtime exception triggers a rollback of the whole transaction.
D	Camouflage	Membership length has no effect on rollback.

12. Answer A Concept 8

An Angular front end served at http://localhost:4200 calls a Spring API at http://localhost:8080. Both run on the same laptop, ~~which has 16 GB of RAM.~~ The browser blocks the call. What fixes it?

Target word: cause and fix of a cross-origin block **Decides:** Different port means a different origin.

A	Correct	The server must send headers that allow that origin.
B	Inverted operation	The server, not the caller, grants cross-origin access.
C	Absolute language	Port is part of the origin.
D	Camouflage	Memory has nothing to do with the origin policy.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	HTTP methods and idempotency	GET reads without side effects; POST creates and is not idempotent; PUT replaces and is idempotent; DELETE removes and is idempotent.	Using GET to change data, or calling POST idempotent.	"Safe to retry" or "same result if sent twice".
2	HTTP status codes	2xx success (201 Created for a new resource), 4xx client error (400 bad input, 401 not authenticated, 403 not allowed, 404 not found), 5xx server error.	Returning 200 for everything, or mixing up 401 and 403.	A described outcome and a question about which code to return.
3	REST resource design	URIs name resources with nouns; the HTTP method carries the action; path variables identify one resource, query parameters filter a collection.	Verbs in URIs such as /getCustomer, or IDs placed in query strings for single resources.	Choosing between endpoint paths.
4	Entity relationships and foreign keys	In a one-to-many relationship the foreign key lives in the table on the many side.	Placing the foreign key on the one side.	"Each X has many Y" and a question about where the key goes.
5	Repository pattern	A repository interface handles data access for one entity; the framework generates standard CRUD and derived queries from method names.	Writing SQL in controllers, or thinking every query needs hand-written SQL.	A method name like findByLastName.
6	Layered architecture	Controller handles HTTP, service holds business logic, repository handles persistence; each layer calls only the one below.	Putting business rules in the controller or the repository.	Where a calculation or rule should live.
7	Transactions	A transaction makes several database writes all-or-nothing; if one fails, all roll back.	Thinking earlier successful writes stay saved after a later one fails.	Two or more related writes, one of which fails.
8	Cross-origin requests (CORS)	A browser blocks calls to a different origin (scheme, host, or port) unless the server sends CORS headers allowing that origin.	Thinking the same host on a different port is the same origin.	A front end on one port calling an API on another.

Course scope (public description): Back-End Programming introduces students to creating back-end components of a web application with the support of framework packages. This course also teaches students how to implement database functionality in a web application and how to create web services. This course requires intermediate expertise in object-oriented programming and the Java language.